YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 1 OF 10 · 40 MINUTES

What do scientists do? 科学家做什么?

Scientists separate what they see (observations) from what they think it means (inferences), and change their story when new evidence appears.

- I can tell the difference between an observation and an inference.
- I can change my inference when I get new evidence.

scientist 科学家 · observation 观察 · inference 推断 · evidence 证据 · clue 线索 · footprint 脚印 · track 足迹

THE LESSON

Skills check	15	Seat groups by plan; hand out skills check and read stems aloud. Silent, alone, no help.
Survey and Today slide	3	Collect checks; read survey aloud, students tick faces. Today slide (1 min): goals, thumbs.
Hook	3	Stage 1 snow tracks: 'Who walked here?' Pair guesses in any language; introduce the series.
Learn	4	Melted snowman: observation or inference? Thumbs on 4 statements; model the two frames.
Do	10	Footprint puzzle in groups: Stage 2 at about 3 min, Stage 3 at about 6. Ask 'How do you know?'
Think and Reflect	5	Two Reporters share; 'Can we be 100% sure?'; exit ticket; first logbook line; collect both.

Hands-on: Footprint puzzle (paper): groups open 3 reveal cards stage by stage, recording observations and inferences.

Collect or assess: Baseline skills check (out of 20) and interest survey into the tracker for Week 10; exit ticket; first logbook page.

SAFETY

- No practical hazards beyond normal classroom rules.
- Keep bags and coats under the tables so walkways stay clear.

WATCH OUT

- Skills check: full 15 minutes, no help; never go through or return papers (same paper in Week 10).
- An inference is not a guess: link it to an observation with 'because'. Short of time? Skip Stage 2.

Language: Frames: 'I observe ___.' 'I infer ___ because ___.' Chinese talk welcome; report in English. 推断 is not 猜 (guess).

Next week: Week 2 Do Now: mystery cups. This week make 12 cups, print and cut question sort cards, find a hardback book.

PREP FOR ONE CLASS

before the lesson about 187 min (154 could be done by someone else); after about 108 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order Week 1 kit: A4 card, envelopes, pouches, A3, name cards, pencils, box 15 min · technician
- [ ] Book reprographics: 24 logbooks, 28 dictionaries (A5, 90 gsm) 10 min · teacher

THE WEEK BEFORE

- [ ] Print, laminate and cut role cards: 6 sets of 6 (6 A4 pouches) 25 min · technician
- [ ] Print 30 language cards double-sided and laminate (30 A5 pouches) 30 min · technician
- [ ] Print 18 puzzle cards on card, skills checks, surveys; make 24 name cards 30 min · technician
- [ ] Print 24 worksheets double-sided; check exit ticket cut line first 5 min · reprographics
- [ ] Write seating plan, set up tracker tab, print mark scheme and kit card 23 min · teacher
- [ ] Optional, if not using card: print puzzle cards on paper, laminate (18 A4) optional 20 min · technician
- [ ] Put up 3 A3 posters and A4 frame poster; print, cut 6 sets of frame cards 35 min · technician

THE DAY BEFORE

- [ ] Label 18 stage envelopes, 1 puzzle card in each; Stages 2 and 3 for desk 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out name cards, role cards, pencils; papers and envelopes on desk 8 min · technician

AFTER THE LESSON

- [ ] Check envelope sets; enter scores in tracker; file papers, do not return 28 min · technician
- [ ] Read exit tickets; mark skills checks out of 20 within the week 80 min · teacher
- [ ] Optional: enter question-by-question marks on Question_analysis tab optional 15 min · technician

PRINT

- Skills check x24 (3 sheets each, double-sided, last side blank)
- Interest survey x24 (half-A4, 2 per A4 sheet, cut in half: 12 sheets)
- Worksheet x24 (2 sheets each, double-sided)
- Teacher copies: kit card, skills check mark scheme (2 pages)
- Puzzle cards x6 sets of 3 (18 A4 card, teacher note on back of Card 3)
- Role cards x6 sets (1 A4 sheet each, laminate and cut)
- Classroom language card x30 (A5, double-sided, laminate)
- Posters: 3 A3 language posters, 1 A4 sentence frame poster
- Sentence frame cards x6 sets of 16 (24 A4 sheets, cut)
- Reprographics booklets: logbook x24, dictionary x28 (A5)

GET OR MAKE (NEW THIS LESSON)

- A4 white card x18 sheets (puzzle cards)
- C4 envelopes x18 (puzzle stages)
- A4 laminating pouches x6 (role cards)
- A5 laminating pouches x30 (language cards)
- A4 pouches x18 for paper puzzle cards (optional)
- A3 paper x3 sheets (posters)
- Name cards x24
- HB pencils x24, plus spares on the teacher desk
- Class box with lid x1
- Class assessment folder x1 (if not already there)
- Laminator and guillotine or scissors (access)
- Rulers for the tables (optional)
- Puzzle envelope sets x6 (make)
- Group seating plan with starting roles (make)

Have ready from earlier weeks: Nothing.